

MinMaxEntropy: Bound Model Errors for Side-Channel Leakages From Information Theory

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Abstract—Side-channel attacks and evaluations have been incessantly pursuing an accurate leakage model and try to address the following question: “How good is my leakage model?” However, the existing works do not well alleviate the attackers and evaluators from model assumption error and estimation error. The recent work named maximum entropy distribution (MED) model does not depend on any assumptions but uses nonlinear programming Newton–Raphson method to fit the leakage distribution, thus avoiding assumption error and making the estimation error arbitrarily small. It tries to address a more fundamental problem: “How to achieve the optimal leakage model?” but still have to face with two issues: 1) the large deviation of MED model from leakage distribution and 2) the difficulty in determining the moments required in model profiling. In this article, we first introduce the nonlinear programming optimizations Levenberg–Marquardt and Conjugate Gradient methods to tackle the first issue. We then exploit Hopfield neural network to solve the minimum entropy for leakage model. Unlike the MED indicating the theoretically most unbiased, objective and reasonable leakage model, the minimum entropy corresponds to the theoretically most biased, subjective and unreasonable leakage model. This facilitates us to build a MinMaxEntropy bound from the maximum entropy and minimum entropy for estimation errors in leakage model, which theoretically represents the amount of information contained on unused higher moments. This bound well provides theoretical support for the moments constraints required to profile the MED model, thus well tackling the second issue. Experimental results fully demonstrate the superiority of our above schemes.

Index Terms—Information theory, leakage model, maximum entropy, side-channel evaluation.

I. INTRODUCTION

SIDE-CHANNEL attacks, which aim to extract secret information (e.g., power consumption [18], electromagnetic radiation [23], timing [17], acoustic [12], and caching pattern [19]), that is, unintentionally leaked in a cryptographic implementation, have been regarded as one of the most serious threats against cryptographic hard-wares. Among them, power (consumption) attacks are the earliest and most mature technology, and the corresponding evaluations have become extremely important. The evaluations can be classified into three categories: 1) attack test; 2) general evaluation; and 3) formal verification. The latter two focus on evaluating whether the side channel information leakage exists. The first one is the most popular way, and it aims to evaluate the attacker’s ability of key recovery. It exploits a model to approximate the leakage of the intermediate values, and exploits a statistical analysis tool [e.g., Pearson correlation coefficient [3] in correlation power analysis (CPA)] to extract the side-channel information. The leakage model can be a hypothetical one or a profiled one. The former usually happens in nonprofiled attacks like CPA, and hypothetical models like Hamming distance and Hamming weight [3], are used. The latter usually happens in profiled attacks like Template Attacks (TA) [5] and stochastic models [25] as stated in [30], and models like probability density distribution, are used.

A good leakage model has a significant impact on the effectiveness of evaluations and attacks. The better the leakage model, the more information we extract, and the more accurate the security evaluations. In this case, how to profile an accurate or the optimal leakage model continues to be an active area of research. In this article, we pursue an unbiased, objective and reasonable model for the leakages, and try to better address the question: “How to achieve a optimal leakage model?” by measuring the amount of information contained in moment constraints. Related works will be discussed in Section I-A before describing our contributions.

A. Related Works

A perfect leakage model can fully extract the leakage information, thus accurately evaluating the security of the cryptographic implementation against side channel attacks. In view of the importance of the model, existing works strove to pursue an accurate model. The deviation of model comes from

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both model *assumption error* and *estimation error*. The former comes from the assumptions made on the leakage model, and the latter comes from the estimation error of model parameters. The most of profiled attacks assume that the leakages are normally distributed [10]. However, this assumption is too simple, and it is often not enough to use only two parameters (i.e., the mean and variance) to describe the leakage. In other words, the leakage may not well follow this assumption, and assumption error is introduced in this case.

To reduce the assumption and estimation errors, Flament et al. discussed probability density function estimation, and compared parametric estimation and histogram estimation in [11], but did not consider information on higher-order moments. The leakage may not well follow this as mentioned above. Therefore, some works like [4], [13], described the leakage characteristics by using information on higher-order moments. However, the corresponding probability density functions were seldom discussed and have not yet given. Several works in leakage detections exploited the hypothesis tests like Welch's t-test [7], correlation ρ -test [8], and χ^2 -test [21], to bound the errors for a bounded model [16]. Other works like [6], [24], and [26], even performed leakage detection and assessment before cryptographic algorithms were implemented on devices. Leakage assessments here seek a standard approach that enables a fast, reliable and robust evaluation of the side-channel vulnerability of the given devices. However, a bounded moment model rather than a probability density distribution leakage model is provided, and it is difficult to launch an attack with such a model. Moreover, all above works only considered no more than the first 4 moments and ignored information on higher-order moments.

Leakage certifications in [10] had a deeper discussion on assumption and estimation errors, and attempted to solve this fundamental problem for side-channel evaluations. This work was further improved in [9]. These above two works tried to better address the question: "*How good is my leakage model?*" It is noteworthy that leakage certification is still directly performed on moments, rather than profiles a probability density distribution model with information from higher-order moments. Moreover, leakage certification still needs to contend with assumption errors and estimation errors under some assumed leakage models like Gaussian, since the leakage may not well follow such distribution as we have mentioned before.

Benefiting from that maximum entropy distribution theoretically corresponds to the most objective, unbiased and reasonable model as described in [15], [28], and [29], the Maximum Entropy Distribution (MED) leakage model was recently proposed in [22], which provided the first probability density distribution leakage model with information on higher-order moments. MED model does not make any assumption on the leakage distribution, but exploits nonlinear programming Newton–Raphson method to find a probability density function to fit the leakage distribution. Therefore, MED model is without assumption error. Moreover, it can theoretically exploit any higher-order moment information to make the estimation error arbitrarily small. However, the MED model with Newton–Raphson method in [22] usually faces with large deviation of MED model from leakage distribution.

Above all, side-channel attacks and evaluations have been incessantly pursuing a good leakage model, but the existing works do not well alleviate the attackers and evaluators from model assumption errors and estimation errors. Although several works have discussed the probability density distribution leakage model under higher-order moments, only the practical MED model was given. Moreover, we still have to face with the deviation from the leakage distribution in MED model, and the difficulty on determining the order of moments required in its profiling. There are still some fundamental problems to be solved on the way to achieve the theoretically most unbiased, objective and reasonable leakage model.

B. Our Contributions

This article aims to overcome the assumption error and estimation error in leakage model profiling, and provide an unbiased, objective, reasonable model with higher-order moments information for attacks and evaluations. We first introduce the nonlinear programming optimizations Levenberg–Marquardt method and Conjugate Gradient method to solve the MED model. This avoids the large deviation of MED model from leakage distribution in Newton–Raphson method under the higher-order moments constraints, so as to obtain a significantly superior and stable model. Furthermore, we exploit Hopfield neural network to estimate the minimum entropy of the leakage samples. Unlike the maximum entropy model indicating the theoretically most unbiased, objective and reasonable leakage model, the minimum entropy indicates the theoretically most biased, subjective and unreasonable leakage density in attacks and evaluations. Finally, we combine the maximum entropy and minimum entropy, and provide a MinMaxEntropy bound for the estimation errors in leakage model, which theoretically represents the amount of information contained on the unused higher-order moments. This bound well conquers the difficulty in determining the moments required in profiling the MED model. Unlike the models that are used to carry out power analysis attacks, such as Hamming weight model, Hamming distance model, polynomial model, etc., MED model is used to help attackers and evaluators estimate the amount of information contained in leakage samples. Therefore, MED model is suitable to make leakage certification and bound the model errors caused by lacking higher-order moments. Experimental results fully verify the superiority of our above schemes.

C. Organization

The remainder of this article is organized as follows. The information entropy, leakage model deviation, moment constraints and the extremal information entropy under given moment constraints are given in Section II. Maximum entropy model estimation, including its principle and our two new nonlinear programming optimizations Levenberg–Marquardt method and conjugate gradient method, are introduced in Section III. Minimum entropy model estimation and its solution Hopfield network are introduced in Section IV, and the MinMaxEntropy bounds for leakage model is provided in

Section V. Experiments are performed on an ATmega328p micro-controller in Section VI and the SASEBO-GII FPGA board in Section VII to demonstrate its practicability and efficiency. Finally, we conclude this article in Section VIII.

II. PRELIMINARIES

A. Information Entropy

Information entropy, the average amount of information after eliminating redundancy, is a very important concept in information theory. Let X denote a discrete random variable consisting of n observations of $\mathbf{x} = (x_1, x_2, \dots, x_n)$, and $\mathbf{p} = (p(x_1), p(x_2), \dots, p(x_n))$ denote their corresponding probabilities. The self-information of a possible event x_i in random event X can be defined as

$$H(x_i) = \log \frac{1}{p(x_i)} = -\log p(x_i). \quad (1)$$

The smaller the probability of its occurrence, the greater the uncertainty and the greater the amount of information contained (i.e., the greater the value of self-information). Shannon defined information entropy (or uncertainty) as

$$H(x) = -\sum_{i=1}^n p(x_i) \ln p(x_i) \quad (2)$$

in [27], which denotes the amount of information contained for all possible events x_i ($1 \leq i \leq n$), i.e., the averaged self-information. If X is a continuous random variable, then the Shannon entropy is

$$H(x) = -\int_a^b f(x) \ln f(x) dx. \quad (3)$$

Here, $[a, b]$ is the integral interval and $f(x)$ is the probability density function.

B. Model Deviation

Power side-channel attacks and evaluations pursue a perfect model to fully extract information from the leakage measurements. However, it is very difficult to achieve this goal and the leakage model is never so perfect, since errors will come from assumptions and estimations.

Assumption Error: Side-channel attacks and evaluations always assume a leakage model, and exploit distinguishers or leakage test tools with it to extract side-channel information from leakages. For example, Gaussian model, including mean and variance, can be used in TA or Welch's t-test. These models play a very important role in this case, and a good model should reflect the leakage as well as possible. However, this goal is hard to achieve, since these models often contain assumption error from subjective assumptions.

Estimation Error: Another kind of error in leakage model estimation comes from parameter deviations.

An example for assumption and estimation errors is given in Fig. 1. The estimation error here happens on the red distribution function and assumption error happens on the yellow distribution function, since it is seriously biased. Estimation error can be made arbitrarily small by exploiting more measurements and using cross validation techniques as

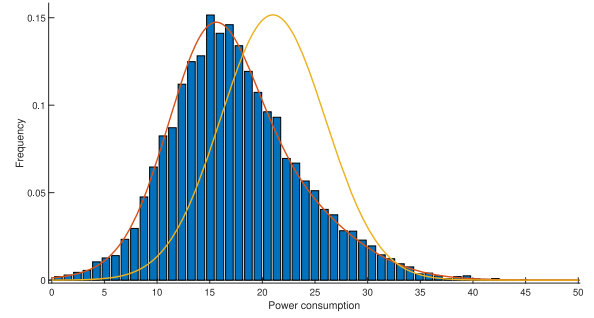


Fig. 1. Assumption and estimation errors in leakage model profiling.

described in [9]. However, assumption error exists in the most of the existing leakage models. We can quantify the goodness-of-fit of the model and the leakage by hypothesis test like Kolmogorov–Smirnov test in [20] and [22].

C. Moment Constraints

Joint information from multiple moments can reflect a probability density distribution much better. The k th order origin moment of a discrete random variable X , μ_k , is defined as

$$\mu_k = \mathbb{E}(x^k) = \sum_i x_i^k p(x_i) \quad (4)$$

which measures the center of the distribution. Here, “ $\mathbb{E}$ ” is the expectation operator. The k th order central moment of random variable X , v_k , is defined as

$$v_k = \mathbb{E}[(X - \mathbb{E}(X))^k] = \sum_i (x_i - \bar{x})^k p(x_i). \quad (5)$$

For example, the first-order central moment v_1 is 0, second-order central moment v_2 is the *variance*, the third-order central moment v_3 is the *skewness*, and the fourth-order central moment v_4 is the *kurtosis*. There is a numerical relationship between the origin moment μ_k and central moment v_k

$$\begin{cases} v_2 = \mu_2 - \mu_1^2 \\ v_3 = \mu_3 - 3\mu_1\mu_2 + 2\mu_1^3 \end{cases} \quad (6)$$

and the computation of origin moment is more convenient when dealing with the probability density estimation of leakages.

It is noteworthy that the higher the order of information moments, the stronger the constraints, and the fewer probability density distribution models that meet the given conditions. In other words, with the more moments information provided, the fewer probability density distribution models meet the conditions and the more determined of the leakage distribution. For example, many probability distribution models satisfying the first two moments constraints cannot satisfy the third-order moment constraint. Therefore, to profile more accurate leakage model, we can make full use of information from higher-order moments in this article, and they reflect the information leaked on different perspectives.

D. Extremal Entropy Under Given Moment Constraints

There are many probability density distributions for leakages that meet the given moment constraints, among which the most *subjective* one corresponds to the minimum entropy, and the most *objective* one corresponds to the maximum entropy. For example, if there is only one possible case for random event X (i.e., $p(x_1) = 1$) in (2), the minimum value of entropy quantified by Shannon entropy is 0. This means, no uncertainty for event X and the amount of information contained is 0. The entropy reaches its maximum $\log n$ if the occurrence probabilities for all n cases of the random event X are equal, i.e., $p(x_1) = p(x_2) = \dots = p(x_n) = (1/n)$. Another example is, the most reasonable assumption before we roll a dice is that, the probability of each side i ($1 \leq i \leq 6$) of the dice facing up is $p(x_i) = 1/6$, which corresponds to a uniform distribution. The entropy reaches $-\sum_{i=1}^6 p(x_i) \ln p(x_i) = -\sum_{i=1}^6 (1/6) \ln (1/6) = \ln 6$ under the natural constraint $\sum_{i=1}^6 p(x_i) = 1$. However, the most unreasonable assumption is that, a side of this dice faces up with a probability of 1.00. The entropy drops to 0.00 under the natural constraint $\sum_{i=1}^6 p(x_i) = 1$ in this case.

We can express the entropy under given moment constraints as follows:

$$H = -\sum_{i=1}^n p(x_i) \log p(x_i) - \sum_{k=0}^K \lambda_k \left(\sum_{i=1}^n x_i^k p(x_i) - \mu_k \right) \quad (7)$$

by exploiting the Lagrange multiplier method and introducing parameter $\lambda = [\lambda_0, \lambda_1, \dots, \lambda_K]^\top$. This facilitates the computation of their extremal values. Here, k ($0 \leq k \leq K$) is the order of the moment constraints, “ $\top$ ” denotes matrix transposition, and μ_k is the k th order moment constraint computed from (5). With the above brief introduction, we will further solve the extreme value problems under the constraint conditions in the next Sections III and IV.

III. MAXIMUM ENTROPY DISTRIBUTION ESTIMATION

Nonlinear programming optimization was introduced to solve maximum entropy in [22], but this faced with the large deviation of MED model from leakage distribution with a high probability when introducing information from higher-order moments. Before we seek a continuous fitting function to better describe the leakage, we should understand that the side channel leakages we sampled are “discrete,” but the probability density distribution is assumed to be “continuous.” Here, we will introduce the maximum entropy model and its two solutions Levenberg-Marquardt method and Conjugate Gradient method in this section, which significantly improve the probability of model convergence or even guarantee the convergence.

A. Maximum Entropy Model

We first formalize the problem of solving the maximum entropy as follows:

$$\begin{aligned} \max : H(x) = & -\sum_{i=1}^n p(x_i) \log p(x_i) \\ & - \sum_{k=0}^K \left(\lambda_k \sum_{i=1}^n x_i^k p(x_i) - \mu_k \right). \end{aligned} \quad (8)$$

The above (8) is constrained by the moments μ_k -s computed from leakage samples

$$\mu_k = \sum_{i=1}^n x_i^k p(x_i) \quad (9)$$

as described in Section II. Here, $0 \leq k \leq K$, and the first K moments are exploited here. Here λ_k is the k th order element of the unknown Lagrange multipliers $\lambda = [\lambda_0, \lambda_1, \dots, \lambda_K]^\top$ as we mentioned before.

Then, by differentiating $H(x)$ w.r.t. $p(x)$, we get

$$\frac{\partial H(x)}{\partial p(x)} = -\sum_{i=1}^n (\log p(x_i) + 1) - \sum_{k=0}^K \lambda_k \left(\sum_{i=1}^n x_i^k \right). \quad (10)$$

The maximum entropy always appears at the extreme points on function $H(x)$. Therefore, we set $(\partial H(x)/\partial p(x)) = 0$ and get

$$-\sum_{i=1}^n (\log p(x_i) + 1) - \sum_{k=0}^K \lambda_k \left(\sum_{i=1}^n x_i^k \right) = 0. \quad (11)$$

Then we transform (11) and obtain

$$\log p(x_i) = -\sum_{k=0}^K \lambda_k x_i^k - 1. \quad (12)$$

We make $\lambda'_0 = \lambda_0 + 1$ to simplify (12) and get

$$\log p(x_i) = -\sum_{k=1}^K \lambda_k x_i^k - \lambda'_0. \quad (13)$$

Since there is only a numerical difference between λ'_0 and λ_0 , for the sake of simplification, we will not distinguish them in the remaining of this article. In this case, the solution of problem becomes

$$p(x) = \exp \left[-\sum_{k=0}^K \lambda_k x^k \right]. \quad (14)$$

We combine this solution with the constraints and further obtain

$$g_k(\lambda) = \int x^k \exp \left[-\sum_{k=0}^K \lambda_k x^k \right] dx = \mu_k. \quad (15)$$

For convenience, we define y as the probability of the independent random variable x , and the original problem can be transformed into computing the K -dimensional vector $\lambda = [\lambda_0, \lambda_1, \dots, \lambda_K]^\top$ denoting the coefficient of each moment constraint with the probabilities y -s obtained from the leakages. Obviously, y is the frequency of x in the probability density distribution.

Suppose that x_i -s ($1 \leq i \leq n$) are the power consumption values in the middle of the n columns when the probability distribution of leakage is represented by a histogram. We obtain a matrix $\mathbf{X}$ as

$$\mathbf{X} = \begin{bmatrix} 1 & x_1^1 & x_1^2 & \dots & x_1^K \\ 1 & x_2^1 & x_2^2 & \dots & x_2^K \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n^1 & x_n^2 & \dots & x_n^K \end{bmatrix}. \quad (16)$$

We bring λ and $\mathbf{X}$ into (14) and get

$$y(\lambda, x) = \exp[\mathbf{X}\lambda]. \quad (17)$$

Since vectors $x = [x_1, x_2, \dots, x_n]$ and $y = [y_1, y_2, \dots, y_n]$ can be directly computed from the leakages, the maximum entropy problem can be solved as follows:

$$\begin{aligned} \min: f(\lambda, x) &= \frac{1}{2} \sum_{i=1}^n (y_i - Y_i)^2 \\ &= \frac{1}{2} (\mathbf{X}\lambda - Y)^T (\mathbf{X}\lambda - Y) \\ &= \frac{1}{2} \lambda^T \mathbf{X}^T \mathbf{X} \lambda - \lambda^T \mathbf{X}^T Y + \frac{1}{2} Y^T Y \end{aligned} \quad (18)$$

by minimizing the difference between their frequencies and the computed probabilities, and this comes into solving the λ which minimizes $f(\lambda, x)$. Obviously, this can be achieved by computing the partial derivative

$$\frac{\partial f}{\partial \lambda} = \mathbf{X}^T \mathbf{X} \lambda - \mathbf{X}^T Y \quad (19)$$

with respect to λ in (18), since the function reaches the extremum if $(\partial f / \partial \lambda) = 0$. The parameters λ can be computed as follows:

$$\mathbf{X}^T \mathbf{X} \lambda = \mathbf{X}^T Y \quad (20)$$

under $(\partial f / \partial \lambda) = 0$ and (20) can be solved by iterative methods for solving linear equations. Here, we use Levenberg–Marquardt method and Conjugate Gradient method to achieve this goal in Sections III-B and III-C.

B. Levenberg–Marquardt Method

Levenberg Marquardt method, our first scheme to solve the maximum entropy, is a widely used nonlinear least squares algorithm combining the Steepest Descent method and Gauss-Newton extremum method. The difference between the parameters and the optimal solution is relatively large at the beginning of the fitting, and the Steepest Descent method in this synthesis speeds up the iteration. Let $\psi^2(\lambda)$ denote the weighted error square sum between the origin moment g of the leakages and the origin moment $\hat{g}$ computed by the model. It can be expressed as

$$\begin{aligned} \psi^2(\lambda) &= \sum_{i=1}^N \left[\frac{g_i - \hat{g}_i}{\sigma_{gi}} \right]^2 \\ &= (g - \hat{g}(\lambda))^T W (g - \hat{g}(\lambda)) \\ &= g^T W g - 2g^T W \hat{g} + \hat{g}^T W \hat{g}. \end{aligned} \quad (21)$$

Here, σ_{gi} is the measurement error on the i th order origin moment g_i , and the weight matrix W is a diagonal matrix and $w_{ii} = (1/\sigma_{gi}^2)$. Nonlinear programming optimizations should minimize ψ^2 iteratively to find an increment δ of λ so that $\psi^2(\lambda - \delta) \leq \chi^2(\lambda)$. The parameters $\lambda = [\lambda_0, \lambda_1, \dots, \lambda_K]^T$ are very close to their optimal values when the iteration is near the end, and the Gauss-Newton method is used to find the optimal solution accurately.

The gradient of $\psi^2(\lambda)$ to λ in Steepest Descent method is

$$\begin{aligned} \frac{\partial \psi^2}{\partial \lambda} &= 2(g - \hat{g}(\lambda))^T W \frac{\partial}{\partial \lambda} (g - \hat{g}(\lambda)) \\ &= -2(g - \hat{g}(\lambda))^T W \left[\frac{\partial \hat{g}(\lambda)}{\partial \lambda} \right] \\ &= -2(g - \hat{g}(\lambda))^T W J. \end{aligned} \quad (22)$$

Here, J is the Jacobian matrix $[\partial \hat{g}(\lambda) / \partial \lambda]$ with a size of $m \cdot n$, which indicates the local sensitivity of $\hat{g}$ to the parameter λ . The increment δ of the parameter λ in Steepest Descent method is as follows:

$$\delta = \alpha J^T W (g - \hat{g}). \quad (23)$$

Here, α is a positive scalar determining the step length on a direction. Unlike Steepest Descent method, Gauss-Newton method aims to minimize the objective function $\psi^2(\lambda)$. It assumes that the objective function approximates a quadratic function when the parameters approximate the optimal solution. The function with the parameters of the disturbance model under an increment δ can be locally approximated by the first-order Taylor series expansion as

$$\begin{aligned} \hat{g}(\lambda - \delta) &\approx \hat{g}(\lambda) + \left[\frac{\partial \hat{g}(\lambda)}{\partial \lambda} \right] \delta \\ &= \hat{g} + J \delta. \end{aligned} \quad (24)$$

Bring the above approximation (24) into $\psi^2(\lambda + \delta)$ in (21) and we can get

$$\begin{aligned} \psi^2(\lambda + \delta) &\approx g^T W g + \hat{g}^T W \hat{g} - 2g^T W \hat{g} \\ &\quad - 2(g - \hat{g})^T W J \delta + \delta^T J^T W J \delta. \end{aligned} \quad (25)$$

The parameters λ -s in Gauss-Newton method are finally solved as follows:

$$\begin{cases} \frac{\partial}{\partial \delta} \psi^2(\lambda + \delta) \approx -2(g - \hat{g})^T W J + 2\delta^T J^T W J \\ [J^T W J] \delta = J^T W (g - \hat{g}) \end{cases} \quad (26)$$

and making $(\partial \psi^2 / \partial \delta) = 0$. The Steepest Descent method is intuitive and simple, but its convergence is slower compared to Gauss-Newton method.

The Levenberg–Marquardt method flexibly and adaptively updates the increment δ as

$$[J^T W J + \beta I] \delta = J^T W (g - \hat{g}). \quad (27)$$

It updates the increment δ more like Gauss-Newton method when the damping parameter β is small, and more like Steepest Descent method when β is large. It is noteworthy that the initial damping parameter β is generally large, so that the initial update of λ -s happens on the fastest descent direction. With the iterations, the damping parameter β decreases and becomes more stable, the update of λ -s is more like Gauss-Newton method, and Levenberg–Marquardt method finds the local optimal solution more quickly in this case.

C. Conjugate Gradient Method

Although every step in Steepest Descent method is toward the local optimum, it will always choose very approximate directions in different iterations. Every step will not optimize one direction to the end, and error cannot be eliminated by updating the direction and step size once. Moreover, gradient in each step in Steepest Descent method is orthogonal to its previous gradient. Therefore, the path of the updated parameter is serrated. To improve the Steepest Descent method, here, we introduce conjugate gradient (CG) method, another popular iterative method for solving large systems of linear equations. Unlike LM method, if the Conjugate Gradient method selects an optimization direction, it can eliminate the error in one iteration on this direction under the selected step size. In other words, it is unnecessary to update error on this direction in the subsequent optimizations.

Let $\{d_{(0)}, d_{(1)}, \dots, d_{(n)}\}$ denote a set of orthogonal search directions, Conjugate Gradient method takes only one step to finish the optimization on one direction, and all the optimizations can be finished in n steps in this case. Specifically, to make each step finish all optimization on one direction, the search directions should be orthogonal to the rest error e_{i+1} as

$$d_i^\top e_{i+1} = 0. \quad (28)$$

Next, we determine the step size $a_{(i)}$. Every step in CG makes the result change

$$x_{i+1} = x_i + \alpha_i d_i. \quad (29)$$

Then, we bring (29) into (28) and get

$$\alpha_i = -\frac{d_i^\top e_i}{d_i^\top d_i}. \quad (30)$$

Here, $e_i = x_i - x$. The orthogonality condition e_{i+1} A-orthogonal to d_i is equivalent to finding the minimum point along the search direction. We bring $e_{i+1}^\top A d_i = 0$ to (30) and get

$$d_i^\top A e_{i+1} = d_i^\top A (e_i + \alpha_i d_i) = 0 \quad (31)$$

and

$$\alpha_i = -\frac{d_i^\top A e_i}{d_i^\top A d_i}. \quad (32)$$

Benefiting from (20), the Conjugate Gradient method can perform as follows.

- 1) First, we initialize λ_0 as an arbitrary value, compute the residual r_0 of linear system $r_0 = X^\top X \lambda_0 - X^\top Y$, and update d_0 as $d_0 = -r_0$.
- 2) Second, the iteration starts and updates parameters as follows: $\alpha_i = -(r_i^\top d_i / d_i^\top X^\top X d_i)$, $\lambda_{i+1} = \lambda_i + \alpha_i d_i$, $r_{i+1} = X^\top X \lambda_{i+1} - X^\top Y$, $\beta_i = (r_{i+1}^\top r_{i+1} / r_i^\top r_i)$, and $d_{i+1} = -r_{i+1} + \beta_i d_i$.
- 3) The iteration ends when residual r_i is small enough or the iteration has run n times.

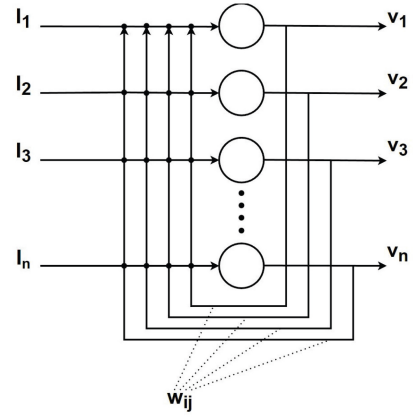


Fig. 2. Hopfield neural network with n inputs and n outputs.

IV. MINIMUM ENTROPY DISTRIBUTION ESTIMATION

Unlike the maximum entropy given in Section III, there may have multiple extremum values for the minimum entropy when profiling the leakage model. Moreover, nonlinear programming strategies like Levenberg–Marquardt method and Conjugate Gradient method, cannot apply to this scenario. Here, we exploit Hopfield neural network [14] to estimate the minimum entropy.

A. Hopfield Neural Network

Hopfield neural network uses the Lyapunov energy function, of which the minimum corresponds to the stationary point of the network. This ensures the convergence to the local minimum. To reach the equilibrium point, the Hopfield neural network would find the minimum of the energy function. A single-layer feedback nonlinear continuous Hopfield neural network with n inputs $\{I_1, I_2, \dots, I_n\}$ is shown in Fig. 2. The input-output relationship of each neuron is a continuously differentiable monotonic increasing function. Specifically, the input of each neuron is a state variable varying with time, and is directly related to the external input and the signals from other neurons. The output of each node is fed back to their inputs. Therefore, the state variable directly affects the inputs, turns the system into a dynamic one that varies with time, and finally converges to the global minimum of the energy system. Obviously, we can consider the entropy in leakage model profiling as the energy function, the moment constraints as the original inputs, and exploit Hopfield neural network to find the estimation of minimum entropy.

As we mentioned above, the state variable directly affects the inputs, and turns the Hopfield neural network system into a dynamic one. Each neuron is composed of an amplifier in this case. The stability of Hopfield network is reflected by its energy function as follows:

$$E = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n w_{ij} v_i v_j - \sum_{i=1}^n v_i I_i. \quad (33)$$

Let u_i denote the state of the i th neuron, the output v_i of the neuron satisfies

$$v_i = f_i(u_i) \quad (34)$$

and its shape is generally S-type (e.g., the *sigmoid* function). If w_{ij} and w_{ji} are the same, i.e., the weight is symmetrical, it can be proved that the derivative of energy function E to time t is non positive. In other words, E decreases monotonically with time t as

$$\frac{\partial E}{\partial t} \leq 0 \quad (35)$$

and $(\partial E/\partial t) = 0$ holds if and only if $(\partial v_i/\partial t) = 0$. Therefore, the whole Hopfield network is always attracted to the minimum of the energy function, and the bounded energy function E can be well proved. We can make use of Hopfield network to construct the corresponding energy function, and obtain the minimum entropy under a fixed moment constraint when its energy reduces and finally becomes stable at the minimum point. The profiling of an accurate leakage model is finished.

B. Solution for the Minimum Entropy

To solve the minimum entropy using Hopfield network, we first need to determine the total energy function. For leakage profiling, the frequency p_i of the leakage distribution is replaced by v_i in Hopfield network given in Section IV-A, thus the information entropy is as follows:

$$H = - \sum_i v_i \log(v_i). \quad (36)$$

The r th moment constraint can be rewritten as

$$J_r = \sum_{i=1}^n g_{ri} v_i - a_r. \quad (37)$$

Here, $g_{ri} = x_i^r$, the original r th order moment constraint $a_r = \sum_{i=1}^n g_{ri} p_i$, and J_r updates in iterations. To guarantee that the energy $E \geq 0$, we can square the moment constraints to make the corresponding term always greater than 0. The whole objective energy function of Hopfield network for solving the minimum entropy can be expressed as

$$E = -A \sum_{i=1}^n v_i \log v_i + \sum_{r=0}^m \lambda_r \left(\sum_{i=1}^n g_{ri} v_i - a_r \right)^2 \quad (38)$$

after introducing the moment constraints into the information entropy function given in (36). Here, A and λ need to be greater than 0.

The total objective function given in (38) includes two parts: 1) the information entropy part from (36) and 2) the difference part between the initial frequencies and the current frequencies in the moment constraints. The second part is 0 in the initial state of the network since $a_r = \sum_{i=1}^n g_{ri} v_i$. The violation of moment constraints in the second part leads to the increase of energy, since with the update of frequency distribution, the whole energy distribution gradually dissatisfies the initial moment constraints. However, the first part continues to decrease in iterations, and leads to the decrease of the whole energy. In other words, the information entropy decreases slowly and reaches the minimum entropy acceptable to the given moment constraints when the energy of the whole Hopfield network finally reaches the minimum. The corresponding parameters v_i -s are the frequencies in

probability density distribution corresponding to the minimum entropy we pursue.

With the help of the above explanation, let us consider the whole objective function E in (36) again. The neuron state updates as follows:

$$\frac{\partial u_i}{\partial t} = A(\log(v_i) + 1) - 2 \sum_{r=0}^m \lambda_r g_{ri} \left(\sum_{j=1}^n g_{ri} v_j - a_r \right). \quad (39)$$

Here, u_i is the state of the i th neuron. The relationship between the neuron state u_i and its output v_i is related by the *sigmoid* function as

$$v_i = f(u_i) = \frac{1}{1 + \exp(-\beta u_i)} \quad (40)$$

as given in (34). Now we prove that the whole energy function E is a monotonically decreasing function of time. We take the derivative of function E with respect to time t , and obtain

$$\begin{aligned} \frac{\partial E}{\partial t} = & -A \left(\sum_{i=1}^n \log(v_i) + n \right) \frac{\partial v_i}{\partial t} \\ & + 2 \sum_{r=0}^m \lambda_r \sum_{i=1}^n g_{ri} \frac{\partial v_i}{\partial t} \left(\sum_{j=1}^n g_{ri} v_j - a_r \right). \end{aligned} \quad (41)$$

The above (41) can be further expressed as

$$\begin{aligned} \frac{\partial E}{\partial t} = & - \sum_{i=1}^n \frac{\partial v_i}{\partial t} (A(\log(v_i) + 1) \\ & - 2 \sum_{r=0}^m \lambda_r g_{ri} (\sum_{j=1}^n g_{ri} v_j - a_r)) \end{aligned} \quad (42)$$

and its second part is the state update equation of neurons, i.e., $(\partial u_i/\partial t)$. Since

$$\frac{\partial E}{\partial t} = - \sum_{i=1}^n \frac{\partial v_i}{\partial t} \frac{\partial u_i}{\partial t} \quad (43)$$

$(\partial v_i/\partial t) = (\partial v_i/\partial u_i)(\partial u_i/\partial t)$, and $(\partial v_i/\partial u_i) = f^{-1}(v_i)$, we obtain

$$\frac{\partial E}{\partial t} = - \sum_{i=1}^n f^{-1}(v_i) \left(\frac{\partial u_i}{\partial t} \right)^2. \quad (44)$$

Here, $f(v_i)$ is the *sigmoid* function, which increases monotonically and globally in the definition domain. Therefore, $f^{-1}(v_i) \geq 0$, we get the derivative of the energy function with respect to time $(\partial E/\partial t) \leq 0$, and $(\partial E/\partial t) = 0$ holds when $(\partial u_i/\partial t) = 0$. The Hopfield network updates u_i and v_i after each round of state change until obtains the minimum entropy.

For complexity, Hopfield neural network can work in parallel, the calculation amount will not increase greatly with the dimension, and Hopfield network can solve the optimization problem on the minimum entropy with polynomial complexity.

V. LEAKAGE BOUNDS FROM THE MAXIMUM AND MINIMUM ENTROPY

Let $S_{\max}(C_k)$ ($S_{\min}(C_k)$) denote the maximum (minimum) entropy acquired from the first k moments $C_k = \{\mu_0, \mu_1, \mu_2, \dots, \mu_k\}$. Here, the μ_1, μ_2, μ_3 and μ_4 correspond

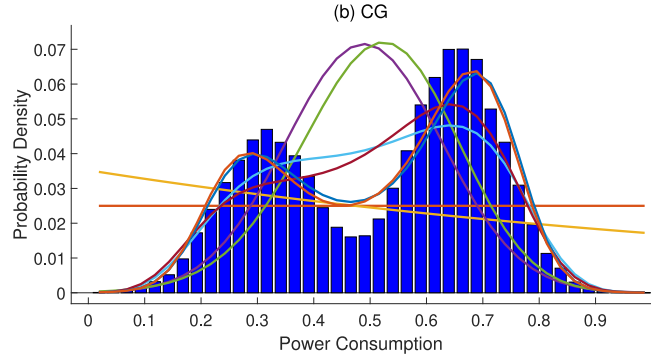
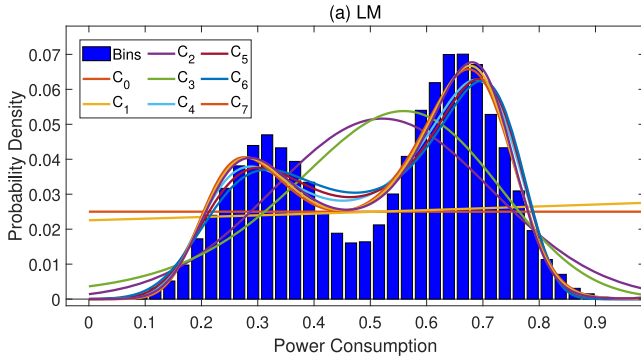


Fig. 3. Fitting performance of LM and CG methods.

the mean, variance, skewness and kurtosis as explained before. Although the distribution is very complex, our two new non-linear programming solutions Levenberg–Marquardt method and Conjugate Gradient method introduced in Section III can still fit it very well (see the example shown in Fig. 3). Here, we have to face with the difficulty in determining the order of moments required in MED model profiling as mentioned in Section I-A. Benefiting from the MinMax information measure in [15], [28], and [29], here, we combine the minimum entropy introduced in Section IV with maximum entropy, and build the MinMaxEntropy bounds for leakage model. This theoretically helps us determine the order of moments required in MED model profiling and solve the second problem described in abstract.

The maximum entropy distribution model corresponds to the theoretically most unbiased, objective and reasonable leakage model, and the minimum entropy distribution model corresponds to the theoretically most biased, subjective and unreasonable case. Their difference (i.e., gap) means the information contained in a moment constraint set as described in [15], [28], and [29], and we named it as MinMaxEntropy bounds for leakage model. An objective, unbiased and accurate leakage model is required in both attacks and evaluations. Obviously, the MED model is the one we pursue. The MinMaxEntropy bounds tell us all the remaining information contained in all the unused higher-order moments, so that we can determine the order of moments required to profile the maximum entropy distribution model.

Let $MME(C_k)$ denote the MinMaxEntropy bounds gap. The maximum entropy distribution model will be uniform if we only consider μ_1 (i.e., the natural constraint). The entropy value, $S_{\max}(C_k)$, reaches $\ln n$ in this case (n is the number of bins in this distribution). The corresponding minimum entropy distribution will be a determined event with the entropy value $S_{\min}(C_0) = 0$. For any constraint set C_k , the MinMaxEntropy gap is evaluated by corresponding minimum and maximum entropy, and is given by

$$MME(C_k) = S_{\max}(C_k) - S_{\min}(C_k). \quad (45)$$

The model has a smaller maximum entropy $S_{\max}$ and a larger minimum entropy value $S_{\min}$ when considering more moments. The narrowed gap in MinMaxEntropy bounds caused by the newly introduced moments represents the

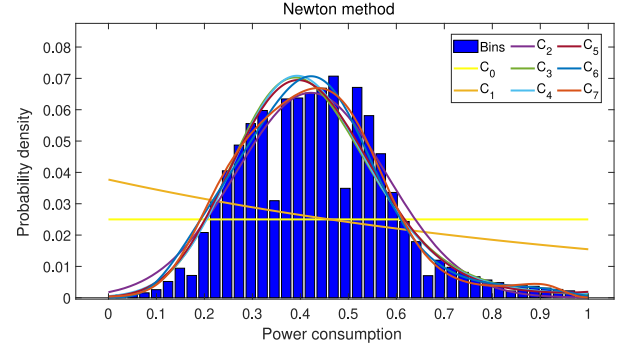


Fig. 4. Fitting performance of Newton–Raphson method.

information contained on them. This gap decreases very slowly or even almost stops decreasing when the moment constraint set reaches a certain degree, and higher moments will not help us profile a more accurate maximum entropy leakage distribution model in this case. Obviously, the MinMaxEntropy bounds can help us figure out how many moments for constraints we need for model profiling.

VI. EXPERIMENTS RESULTS ON ATMEGA328P MICRO-CONTROLLER

A. Experimental Setups

Our first experiments are performed on the AES-128 algorithm (provided by [2]) implemented on ATmega328p micro-controller with a clock of 16 MHz. We randomly encrypt 2 000 000 plaintexts and use a WaveRunner 8104 oscilloscope to acquire power traces. The sampling rate is set to 1 GS/s. The remaining experiments are performed on the MATLAB R2016b on a HP desktop computer with 6 Intel Xeon E5-1650 CPUs, 16-GB RAM and a Windows 10 operating system. We perform CPA on 10 000 measurements to extract a Points of Interest (POI) [8] of the first S-box in the first round with the largest correlation coefficients and profile its leakage model. Before this, its parameter x^k is too large and we transform the range of x into $[0, 1]$ by using $x' = (x - x_{\min}) / (x_{\max} - x_{\min})$, which also reduces singular matrix.

B. Results

Three maximum entropy optimizations, Newton–Raphson, LM, and CG, continuously iterate to achieve better fitting.

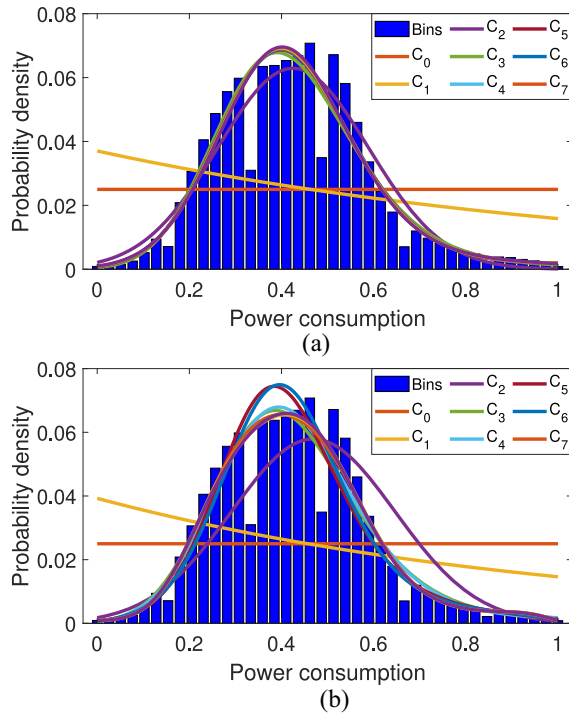


Fig. 5. Fitting performance of LM and CG methods.

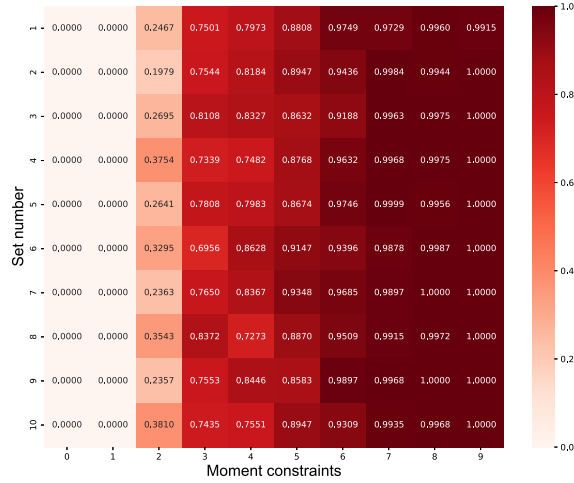


Fig. 6. KS-test results of Newton-Raphson method in cross-validation.

An example is given in Figs. 4 and 5 to illustrate how they update parameters to achieve the final MED model (i.e., MED function). Hundreds or thousands of samples are also possible for leakage profiling. It is noteworthy that the more samples we use, the more stable and accurate of the leakage distribution. Here, 500 000 samples are randomly extracted in each repeated experiment, and we further fix the number of bins to make the comparison fair. Specifically, the number of bins and the iteration precision are set to 40 and 10^{-12} , respectively. Besides, fast convergence requires a suitable way of segmenting the histograms, trying to make the histograms as continuous as possible, thus avoiding that the histograms have empty columns in the middle due to segmentation problems.

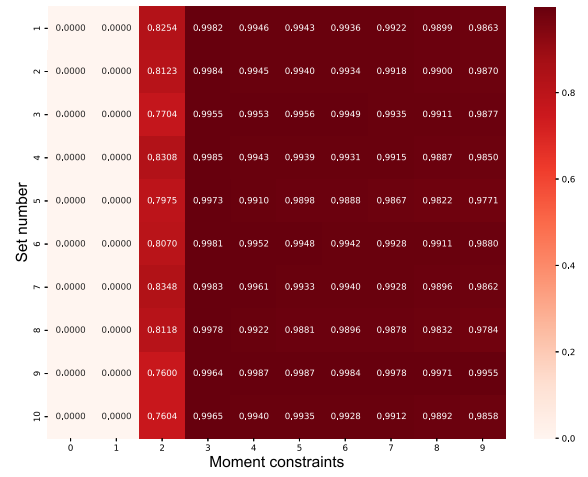


Fig. 7. KS test results of LM method in cross-validation.

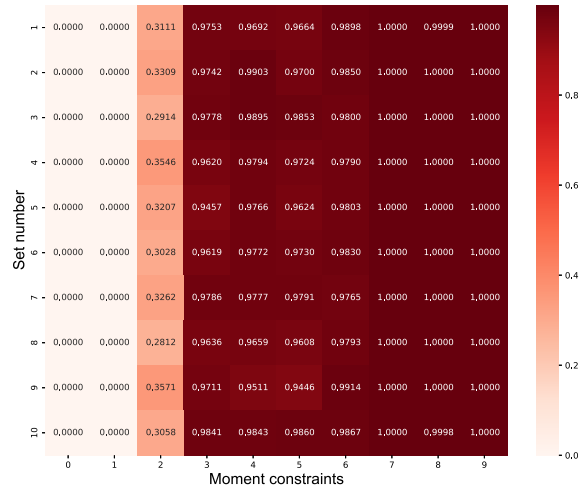


Fig. 8. KS test results of CG method in cross-validation.

This can significantly improve the accuracy and convergence speed of model profiling.

Obviously, the MED model barely conveys any information under constraints like C_0 or C_1 (see definition in Section V). The reason for this phenomenon is that, any probability density will be suitable and a uniform distribution has a maximum entropy under constraint C_0 , while the probability density only has a particular average under constraint C_1 . The MED model fits the frequency distribution better when considering more moments in constraints set. As the number of iterations and the moment information used increase, the three methods Newton-Raphson, LM, and CG, converge rapidly. In this case, we get a MED model with steep left and gentle right.

It is noteworthy that LM and CG provide a significantly higher-accurate MED model compared to Newton-Raphson under 40 bins, the same iteration accuracy 10^{-12} and the same constraint set in our experiments. Specifically, we randomly divided the 500 000 samples into 10 sets to make a 10-fold cross-validation. Nine sets of data are used for profiling and the remaining set is used for performing KS test on the model for each repetition. The results of KS tests in cross-validation are shown in Figs. 6–8. Their superiority is even more obvious

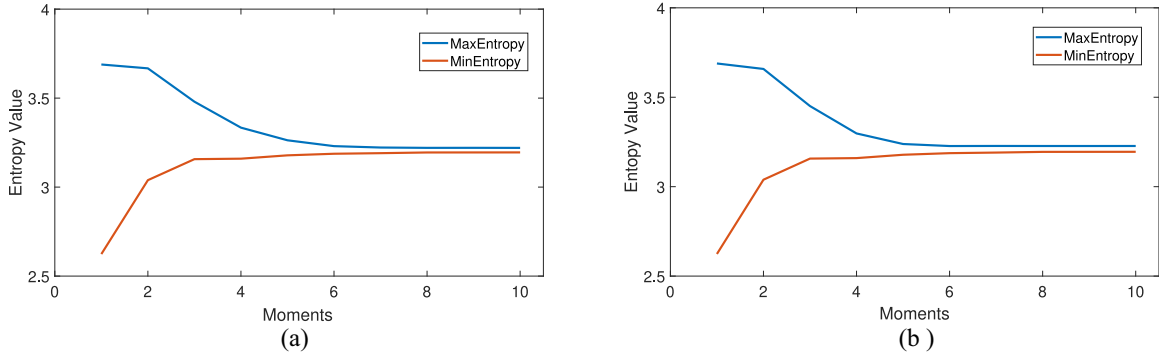


Fig. 9. MaxMinEntropy gap of LM and CG methods.

when the number of bins is small. However, it is still difficult for us to judge how many moments is sufficient to profile an accurate model without considering the minimum entropy. For example, the MED models of Newton–Raphson, LM, and CG only have a tiny change when constraint C_7 (even C_6) is taken into the constraint set. More moment constraints mean a very limited improvement on the fitting accuracy of MED model while bring more computation. Therefore, a sensible way is to find the smallest constraint set while minimize the computation is of great significance. Here, we address this question by using our MinMaxEntropy bound, which can infer us the necessity to put more moments into the constraints set.

The MinMaxEntropy bound reflects the amount of information provided by the constraints, and increases with the constraints set getting expanded, but the higher moments bring less information in our experiments. The averaged MinMaxEntropy bounds for LM and CG combined with minimum entropy are given in Fig. 9. The bound changes little under C_6 or larger constraint sets, and the first 6 or 7 moments may be enough in leakage model profiling. The value of the MinMaxEntropy bound gap are given in Fig. 10. It decreases with more moments constraints brought into the constraints set. This value reflects the amount of rest information that has not been used in MED model. When the constraints set expands to C_6 or C_7 , the gap is closed to 0 and hardly changes with constraints set expanding, which means C_6 or C_7 can provide enough information for leakage profiling, and more moment constraints can hardly make the model better (more accurate). Besides, the MinMaxEntropy bound gap calculated with LM and CG method is lower than Newton method, which infers MED model solved by LM and CG methods extracts more information in higher-order moments constraints than Newton method does.

The MED method can use the information on each given moment constraints to model the leakage. When enough information has been used in the MED model, the MinMaxEntropy bound becomes narrower and hardly changes with more moment constraints brought into profiling. If the bound becomes smooth but is still large, it means probability density distribution function of the target device is complex. The higher moments still contain information but are hard to

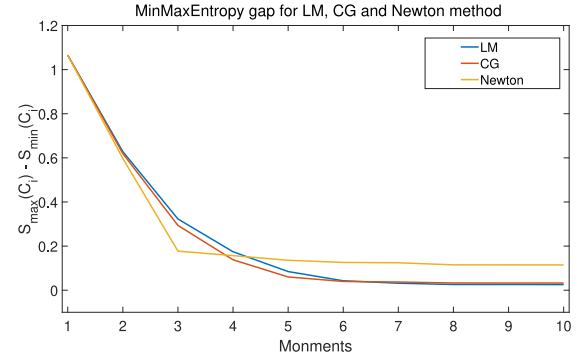


Fig. 10. Entropy bounds of LM, CG, and Newton methods.

be extracted by our MED model, and a larger estimation error will happen in this case.

The MED model solved by Newton–Raphson, LM, and CG converges quickly under higher-order moments, and one more moment brings only an additional neuron to the Hopfield network. Moreover, attackers or evaluators are often willing to invest a lot of their computing power for a sufficiently accurate model, the complexity brought by the above computations is acceptable in this case.

VII. EXPERIMENTS RESULTS ON SASEBO-GII

A. Experimental Setups

To better illustrate performance of our LM and CG methods in MED profiling, we perform the second experiments on the AES_HD dataset from a hardware implementation of AES provided in [1]. It consists of electromagnetic measurements of an unprotected AES-128 implementation on Xilinx Virtex-5 FPGA of a SASEBO-GII evaluation board. The dataset includes 500 000 traces in total with 500 000 randomly generated plaintexts and each trace contains 1250 samples. We select a POI with obvious leakage of the first S-box in the first round, and further normalize it into $[0, 1]$ to reduce the probability of singular matrix. The remaining experiments are performed on the MATLAB R2016b on a HP desktop computer with 6 Intel Xeon E5-1650 CPUs, 16-GB RAM and a Windows 10 operating system, just as detailed in Section VI.

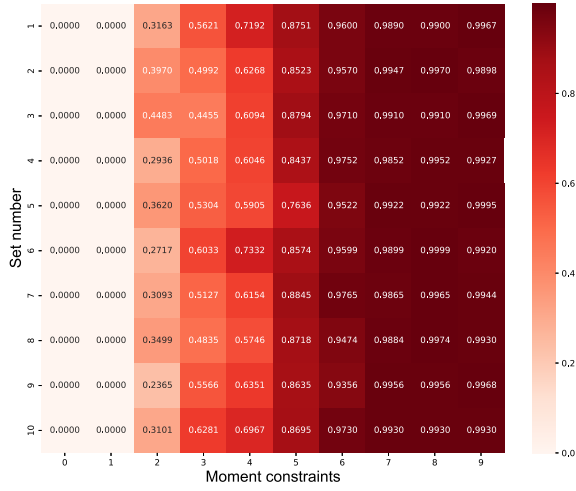


Fig. 11. KS test results of Newton–Raphson method in cross-validation.

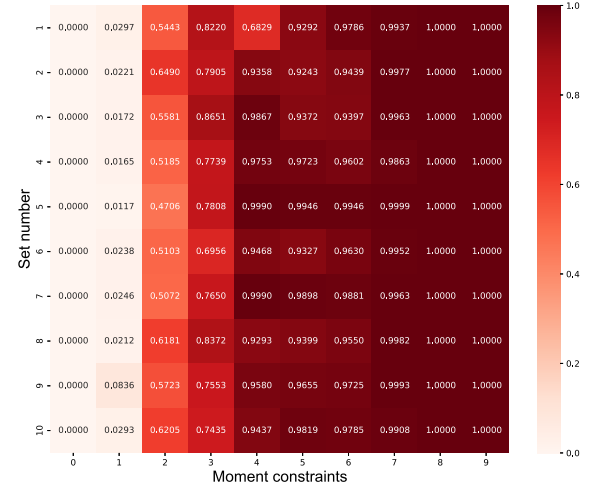


Fig. 13. KS test results of CG method in cross-validation.

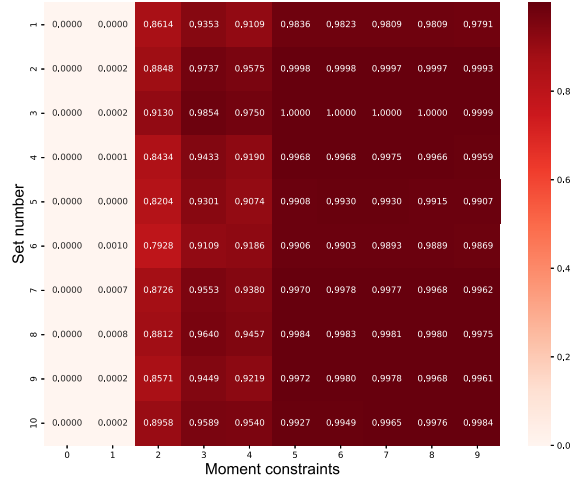


Fig. 12. KS test results of LM method in cross-validation.

B. Results

We randomly extract 300 000 samples and divide them into 10 sets to make a 10-fold cross-validation. The results of the KS tests for Newton–Raphson, LM, and CG methods are displayed in Figs. 11–13. Compared to the results on the dataset acquired from ATmega328p micro-controller given in Section VI, the probability of KS test is a little lower when we use constraint C_3 and constraint C_4 . However, with more moment constraints brought into the constraints set, the performance of MED method becomes very well, our LM and CG methods also work better than Newton–Raphson method in KS tests.

The MinMaxEntropy bounds provided by LM and CG methods are given in Fig. 14. The bound changes very small under C_5 or larger moment constraint sets, and the first 5 or 6 moments may be enough in leakage MED model profiling. The gap becomes particularly narrow and more moment constraints can hardly provide information for MED method to make the model better. Furthermore, the values of the MinMaxEntropy bound gap are given in Fig. 15. It shows LM and CG methods can get a tighter bound than

Newton–Raphson method. This means, we can draw the same conclusion as the one given in Section VI that the MED model solved by LM and CG methods can still extract more information under the same moments constraints than Newton–Raphson method in AES_HD dataset.

VIII. CONCLUSION

An accurate leakage model plays a very important role in side-channel attacks and evaluations. In this article, we introduced Levenberg–Marquardt and Conjugate Gradient methods to tackle the large deviation of MED model from leakage distribution in Newton–Raphson method under the higher-order moments constraints, and gave a MinMaxEntropy bound from the maximum entropy and minimum entropy for estimation errors to conquer the difficulty in determining the moment constraints required in profiling. This bound theoretically represents the amount of information contained on unused higher moments, and well provides theoretical support for the constraints required to profile the MED model. MED model can theoretically exploit information on arbitrary higher moment to profile theoretically most unbiased, objective and reasonable leakage model with the help of MinMaxEntropy bound, thus better address the fundamental problem: “How to achieve the optimal leakage model?”

LM and CG converge very quickly like Newton–Raphson method. LM reports the singular matrix with a smaller probability than Newton–Raphson method, but CG does not report any singular matrix in our experiments. Therefore, we will try to better tackle the singular matrix in both Newton–Raphson method and LM in our future works. MED model can help evaluators and attackers determine the moment constraints required in profiling and evaluating the amount of information in higher-order moments. Besides, masked implementations contain rich higher-order moments information, MED model against them is also an interesting topic, which will be one of our future works. Finally, theoretically quantifying the relationship between the information gain brought by higher-order moments and the attack capability or accuracy of security evaluation is also very interesting.

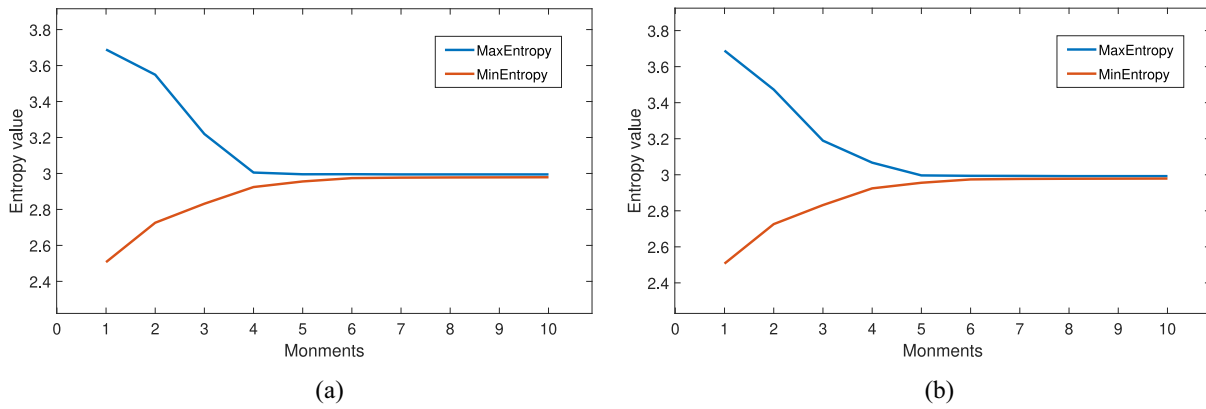


Fig. 14. MaxMinEntropy gap of LM and CG methods. (a) Entropy gap with LM method. (b) Entropy gap with CG method.

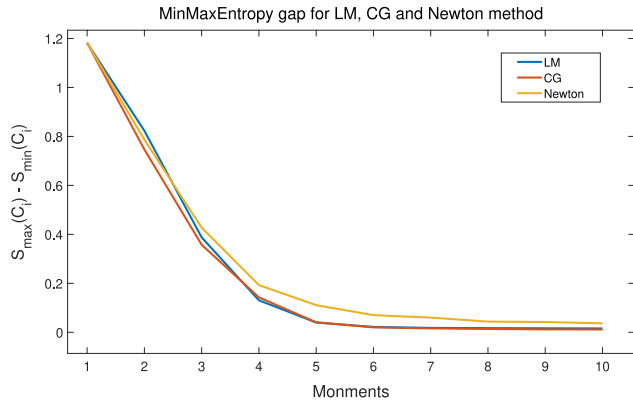


Fig. 15. Entropy bounds of LM, CG, and Newton methods.

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